# AYUSHI SINGH

**PHYSICS EDUCATOR & RESEARCHER • SENIOR SECONDARY & IB-MYP SPECIALIST**

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## PROFESSIONAL SUMMARY

Dynamic and accomplished Physics Educator and Researcher with **9+ years** of pedagogical and research experience delivering premier international curricula (IB Middle Years Programme, High School Physics) and national curricula (Senior Secondary, CBSE, and State Techno) across prestigious institutions. Proven track record in inquiry-based learning, differentiated instruction, laboratory safety, and designing Bloom's Taxonomy-aligned assessments. Demonstrated success in supervising IB Personal Projects with a proven record of students securing the highest grade boundary (Grade 7). Active physics researcher holding an M.Sc. in Physics (Solid State specialization) with published research and hands-on laboratory experience in nanomaterials and metamaterials at defense (DRDO) and university laboratories.

## CERTIFICATIONS & LICENSES

- ManageBac Certified Educator — IB LMS
- Apple Certified Educator — EDTECH
- Certified Microsoft Innovative Educator (MIE) — MICROSOFT

## CORE COMPETENCIES & TECHNICAL SKILLS

**Curriculum & Pedagogy:** IB Middle Years Programme (MYP) Science/Physics, High School Physics, Inquiry-Based Learning, Personal Project Supervision, Bloom's Taxonomy, Differentiated Instruction, Internal Assessments (IA), Lab Safety & Experimentation.

**Educational Technology:** ManageBac LMS, Apple Teacher Tools, Microsoft Innovative Educator Suite, ICT Tool Integration, Digital Assessment Design, Interactive PhET Simulations.

**Scientific Research:** Solid State Physics, ZnO Nanomaterials Synthesis, Metamaterials Characterization, Statistical Modeling, Multivariate Regression, Scientific Documentation.

**Technical Tools:** MATLAB, SPSS, C Language, Turnitin, Microsoft Office Suite (Advanced PowerPoint, Excel, Word).

## PROFESSIONAL TEACHING EXPERIENCE

### MYP Physics Teacher (High School)

07/2024 – Present

Oberoi International School, JVLR Campus — Mumbai, Maharashtra, India

- Plan and execute inquiry-based Physics lessons across high school grade levels aligned with IB MYP curriculum frameworks, conceptual understanding, and learning objectives.
- Implement differentiated instructional strategies to address diverse learning styles, fostering critical thinking, intellectual curiosity, and analytical rigor.
- Guide students through internal assessments, multidisciplinary STEM projects, and hands-on practical investigations adhering strictly to laboratory safety protocols.
- Supervise IB Personal Projects with a proven track record of student success, including achieving the highest grade boundaries (Grade 7).

### MYP Physics & Science Teacher

10/2021 – 06/2024

Lancers International School — Gurugram, Haryana, India

- Instructed Middle Years Programme (MYP) Science across Years 1, 3, 4, and 5, executing inquiry-driven pedagogy aligned with IB criteria and assessment rubrics.
- Served as Community Project Supervisor, mentoring students through interdisciplinary service-learning initiatives and community impact research.
- Supervised Climate Action Projects, guiding scientific inquiry, empirical data collection, and sustainable environmental innovation models.
- Designed formative and summative criteria-based assessments that nurtured conceptual depth, problem-solving skills, and experimental analysis.

### Senior Secondary Physics Teacher

08/2020 – 08/2021

The Shriram School, Aravali — Gurugram / New Delhi, India

- Delivered comprehensive Senior Secondary Physics curriculum to Grades 11 and 12, emphasizing core conceptual mastery and analytical problem-solving.
- Pioneered interactive digital lesson modules, multimedia presentations, and virtual lab simulations to ensure robust student engagement during online instruction.
- Tracked academic progress through standardized diagnostic evaluations and conducted targeted remedial sessions to improve learning outcomes.

## TEACHING EXPERIENCE (CONTINUED)

### Secondary Physics Educator
01/2018 – 08/2020

Johnson Grammar School, Habsiguda — Hyderabad, Telangana, India

- Instructed Physics up to secondary school classes utilizing ICT tools and interactive demonstrations to contextualize complex physical phenomena.
- Formulated question papers, assessment schemes, and holistic lesson plans mapped strictly to Bloom's Taxonomy cognitive domains.
- Conducted rigorous post-evaluation answer analytics to identify systemic misconceptions and implement corrective pedagogical interventions.

### Physics Faculty (Techno Curriculum)
06/2017 – 04/2018

Sri Chaitanya Techno School, Ramanthapur — Hyderabad, Telangana, India

- Delivered rigorous Techno curriculum focused on advanced Physics principles and real-world application mechanisms.
- Coached and mentored secondary students for competitive state and national-level examinations and science olympiads.

### Advanced Physics Specialist (Part-Time)
08/2019 – 06/2024

Johnson Academy — Hyderabad, Telangana, India

- Delivered high-level coaching in advanced Physics, Mathematics, and foundational Chemistry for competitive examinations.
- Facilitated personalized problem-solving strategies, conceptual clarification clinics, and mock test evaluations.

## EDUCATION

### Master of Science (M.Sc.) in Physics
05/2015 – 06/2017

Osmania University — Hyderabad, Telangana, India

- **Academic Standing:** 79% (First Class with Distinction) • **Specialization:** Solid State Physics.
- **Master's Thesis:** *Synthesis and Characterisation of ZnO Nanostructures* – Investigated chemical synthesis pathways and structural/optical properties of zinc oxide nanostructures for optoelectronic and semiconductor applications.

### Bachelor of Science (B.Sc.) in Physical Sciences
05/2012 – 06/2015

University of Delhi — New Delhi, India

- **Academic Standing:** 85% • **Rank 1** in Bachelor's Programme.
- **Core Coursework:** Physics, Chemistry, Mathematics, Botany, Zoology, Visual Basic, and Database Systems.
- **Undergraduate Innovation Project:** Conducted research titled *"A study of crop yield pattern with climate change based on physical parameters: Temperature and rainfall in Western Uttar Pradesh to make future predictions for better crop management and yield."*

## RESEARCH EXPERIENCE & PUBLICATIONS

- **Defence Metallurgical Development Establishment (DMDE / DRDO):** Conducted research on Metamaterials and their characterisation techniques under the mentorship of senior defense scientists.
- **Journal Publication – "Prediction of the Crop Yield" (Delhi University Journal):** Authored predictive study analyzing crop yields across 10 major districts of Uttar Pradesh based on physical parameters (temperature, precipitation, greenhouse gas emissions) using multivariate regression and statistical modeling.

## CONFERENCES & PROFESSIONAL DEVELOPMENT

- **Teach Meet (IB-MYP) (02/2022):** Completed professional development on *Strategies to Succeed in Internal and Summative Assessments.*
- **Indian Institute of Space Science and Technology (IIST), Trivandrum (08/2017):** Participated and presented at the National Conference on space science developments.
- **National Workshop on Material Characterization Techniques (10/2016):** Underwent advanced workshop training in spectroscopic and crystallographic characterization, Hyderabad.
- **World Space Week Conference – ISRO & Spaceadhyaan (06/2016):** Participated in the national conference on astronomical observation and space exploration.
- **"Antardhvani" Industrial Interface Presentation, New Delhi (09/2014):** Presented university innovation research project findings on climatic and crop modeling for industrial utility.

## HONORS & AWARDS

- **Rank 1, B.Sc. Physical Sciences:** Secured first rank across the graduating class at University of Delhi for academic excellence.
- **Undergraduate Research Innovation Grant:** Selected for the prestigious University of Delhi Innovation Research Grant program.

## PROFESSIONAL INTERESTS & CO-CURRICULARS

- **Key Interests:** Experimental Physics Analysis, Condensed Matter Research, Educational Technology (ICT Tools & Simulations), Science Communication, STEM Project Mentorship.